

Internet Appendix for:

Bankruptcy Prediction of Privately Held SMEs Using Feature
Selection Methods

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Appendix IA.1: Estimation techniques

Logistic regressions

When using logistic regressions (LR) as estimation technique, the vector of predicted probabilities for bankruptcy $\hat{\mathbf{y}} = \{\hat{y}_n\}_{n=1,\dots,N} \in [0, 1]^N$ is given by

$$\hat{\mathbf{y}} = \boldsymbol{\iota} \oslash (\boldsymbol{\iota} + \exp(-\mathbf{X}\mathbf{w} - \boldsymbol{\iota}w_0)) \quad (\text{IA.1})$$

where $\mathbf{X} = \{x_{(n,i)}\}_{n=1,\dots,N,i=1,\dots,I}$ is a matrix of values for input variables i derived from the financial statements n , $\mathbf{w} = \{w_i\}_{i=1,\dots,I}$ is a vector of coefficients, w_0 is the intercept coefficient, $\boldsymbol{\iota}$ is an $N \times 1$ vector of ones, and $\oslash$ denotes Hadamard (element-wise) division. For ease of notation, we drop the time indices. We estimate the coefficients $\mathbf{w}$ and w_0 by minimizing the negative of the log-likelihood function $\ell(\mathbf{w}, w_0)$ given by

$$\ell(\mathbf{w}, w_0) = \sum_{n=1}^N [\mathbf{y} \odot (\mathbf{X}\mathbf{w} + \boldsymbol{\iota}w_0) - \log(\boldsymbol{\iota} + \exp(\mathbf{X}\mathbf{w} + \boldsymbol{\iota}w_0))] \quad (\text{IA.2})$$

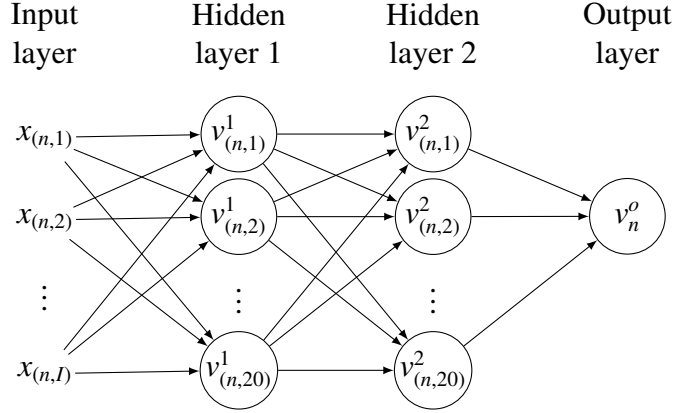
where $\mathbf{y} = \{y_n\}_{n=1,\dots,N} \in \{0, 1\}^N$ is the vector of actual classifications of non-bankrupt (0) or bankrupt (1), and $\odot$ denotes the Hadamard (element-wise) product. For determining the significance of estimated LR coefficients, we use the Wald statistic (Hosmer et al., 2013) to derive their z -scores.

The Deep Artificial Neural Network

We use a feedforward deep artificial neural network (DNN) with one input layer, two hidden layers, and one output layer, as illustrated in Figure IA.A1. The input layer consists of the input variables. The output layer consists of one node representing the output of the DNN, with values between 0 (non-bankrupt) and 1 (bankrupt). The hidden layers each contain twenty nodes. Figure IA.A2 illustrates the computation of the values of the nodes in the hidden and output layers, respectively.

Let $\mathbf{V}^l = \{v_{(n,h)}^l\}_{n=1,\dots,N,h=1,\dots,20}$ be the values of the nodes h for the financial statements n for hidden layer $l \in \{1, 2\}$. The values of the nodes in the first and second hidden layers, respectively,

Figure IA.A1: Illustration of the DNN for any financial statement n



The circles represent nodes, where $v_{(n,h)}^l$ is the value of the nodes $h \in \{1, \dots, 20\}$ for hidden layer $l \in \{1, 2\}$, and v_n^o is the value of the output node. The values are computed as illustrated in Figure IA.A2. All the values of the input variables $\{x_{(n,i)}\}_{i=1, \dots, I}$ are inputs for the computation of values for each of the nodes in the first hidden layer. Further, all the values of the nodes in the first hidden layer are inputs for the computation of values for each of the nodes in the second layer. Finally, all the values of the nodes in the second hidden layer are inputs for the computation of the value of the node in the output layer.

are given by

$$\mathbf{V}^1 = \phi(\mathbf{X}\mathbf{W}^1 + \iota(\mathbf{b}^1)^\top) \quad (\text{IA.3})$$

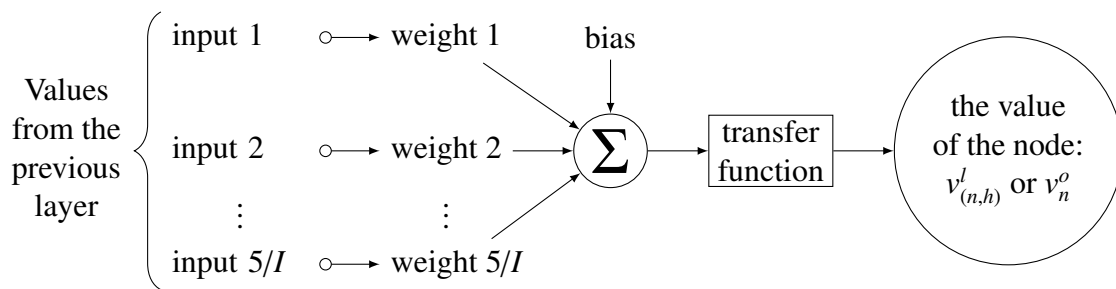
$$\mathbf{V}^2 = \phi(\mathbf{V}^1\mathbf{W}^2 + \iota(\mathbf{b}^2)^\top) \quad (\text{IA.4})$$

where $\mathbf{X} = \{x_{(n,i)}\}_{n=1, \dots, N, i=1, \dots, I}$ is a matrix of values for input variables i , $\mathbf{W}^1 = \{w_{i,h}^1\}$ and $\mathbf{W}^2 = \{w_{h,h}^2\}$ are matrices of weights, $\mathbf{b}^1 = \{b_h^1\}$ and $\mathbf{b}^2 = \{b_h^2\}$ are vectors of biases, $^\top$ denotes the transpose of any matrix or vector, ι is an $N \times 1$ vector of ones, and $\phi(a) = a$ is the linear transfer function. Further, the vector of the nodes in the output layer $\mathbf{v}^o = \{v_n^o\}$ is given by

$$\mathbf{v}^o = \varphi(\mathbf{V}^2\mathbf{w}^o + \iota b^o) \quad (\text{IA.5})$$

where $\mathbf{w}^o = \{w_h^o\}$ is a vector of weights, b^o is a bias value, and $\varphi(a) = \iota \oslash (\iota + \exp(-a))$ is the logistic transfer function.

Figure IA.A2: The computation of the values of the nodes in the hidden and output layers of the DNN



The values of all the nodes (or input variables) in the previous layer are inputs for the computation of each node. First, each of these inputs are multiplied by an associated weight. Second, a sum of all these products and a bias value are the input of a transfer function. Finally, the output of this transfer function is the value of the node.

Appendix IA.2: Evaluation metrics

Average precision (AP) represents the area under the plot of the precision (i.e., the proportion of observations predicted as bankrupt that actually are classified as bankrupt) against the recall (i.e., the proportion of observations classified as bankrupt that are predicted to be bankrupt) across all observations when varying the discrimination threshold. In our setting, the discrimination thresholds vary between 0 and 1, which is the output interval of our models. Further, the area under the receiver operating characteristic curve (AUC) is the area under the plot of the recall against the false positive rate (i.e., the proportion of observations classified as non-bankrupt that are predicted to be bankrupt) across all observations when varying the discrimination threshold. $AUC \in [0.5, 1]$ and $AP \in [0, 1]$, where a higher value indicates a higher explanatory power of the model. Following Hosmer et al. (2013), we consider $AUC \in [0.7, 0.8)$ acceptable, $AUC \in [0.8, 0.9)$ excellent, and $AUC \geq 0.9$ outstanding.

To assess the statistical significance of differences in in-sample fit as well as out-of-sample performance between variable sets, we use the DeLong test, introduced by DeLong et al. (1988), with the improved algorithm of Sun and Xu (2014) for faster implementation. This test evaluates the statistical significance of the differences between two AUCs from distinct models applied to the same dataset.

The Brier (1950) score can be thought of as a cost function. Specifically, the Brier score measures the mean squared difference between *i*) the predicted probability assigned to the possible outcomes for a set of predictors and *ii*) the actual outcome. Hence, the lower the Brier score for a set of predictors, the better is the prediction calibrated.

The Akaike (1974) information criterion (AIC) estimates the relative amount of information lost by a given model. The less information a model loses, the higher the quality of the model. AIC considers the trade-off between the goodness of fit of a model and the simplicity of the model. Specifically, AIC takes into account the tradeoff between bias and variance by introducing a penalty term proportional to the number of input variables. As with the Brier score, a lower AIC value indicates a better prediction.

McFadden's (1974) pseudo-R squared (R^2) is given by $R^2 = 1 - \frac{\ell(\mathbf{w}, w_0)}{\ell(w_0)} \in [0, 1]$ where $\ell(w_0)$ is the log likelihood of a model containing no input variables but only the intercept coefficient w_0 . Hence, higher values of the pseudo-R squared indicate a better model performance.

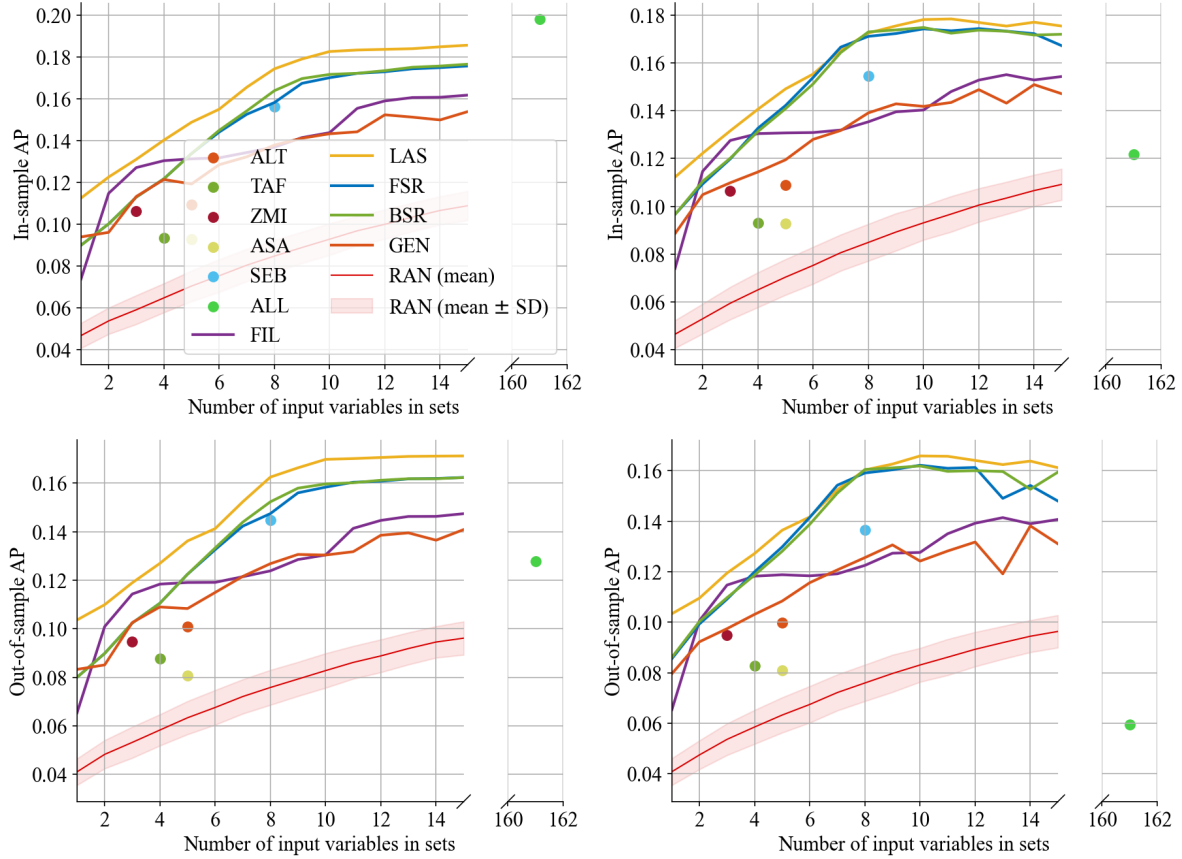
When using decile rankings for evaluating the performance of our bankruptcy prediction models, we group all financial statements used for evaluating our models into deciles based on the estimated probability of bankruptcy. Financial statements that are predicted by the model to have the highest probability of bankruptcy are sorted into the first decile and statements predicted to have the lowest probability of bankruptcy are sorted into decile 10. Model performance is then evaluated by considering the percentage of financial statements that went bankrupt within each decile.

Appendix IA.3: Bankruptcy prediction horizon, extending one year

We test the robustness of our findings using a bankruptcy prediction horizon that extends beyond one year. Specifically, we predict bankruptcy two and three years ahead. We adhere to our categorization of bankruptcy as detailed in Section 3.2 of the manuscript, however, we classify not only the last financial statement but also the previous two as bankrupt. Specifically, when predicting bankruptcy two years ahead, we classify the financial statements as bankrupt if they are either the last or second last financial statement of a company that has filed for bankruptcy within three years from the balance sheet date of its last financial statement. Similarly, when predicting bankruptcy three years ahead, we classify the financial statements as bankrupt if they are among the last, second last, or third last financial statements of a company that has filed for bankruptcy within three years from the balance sheet date of its last financial statement.

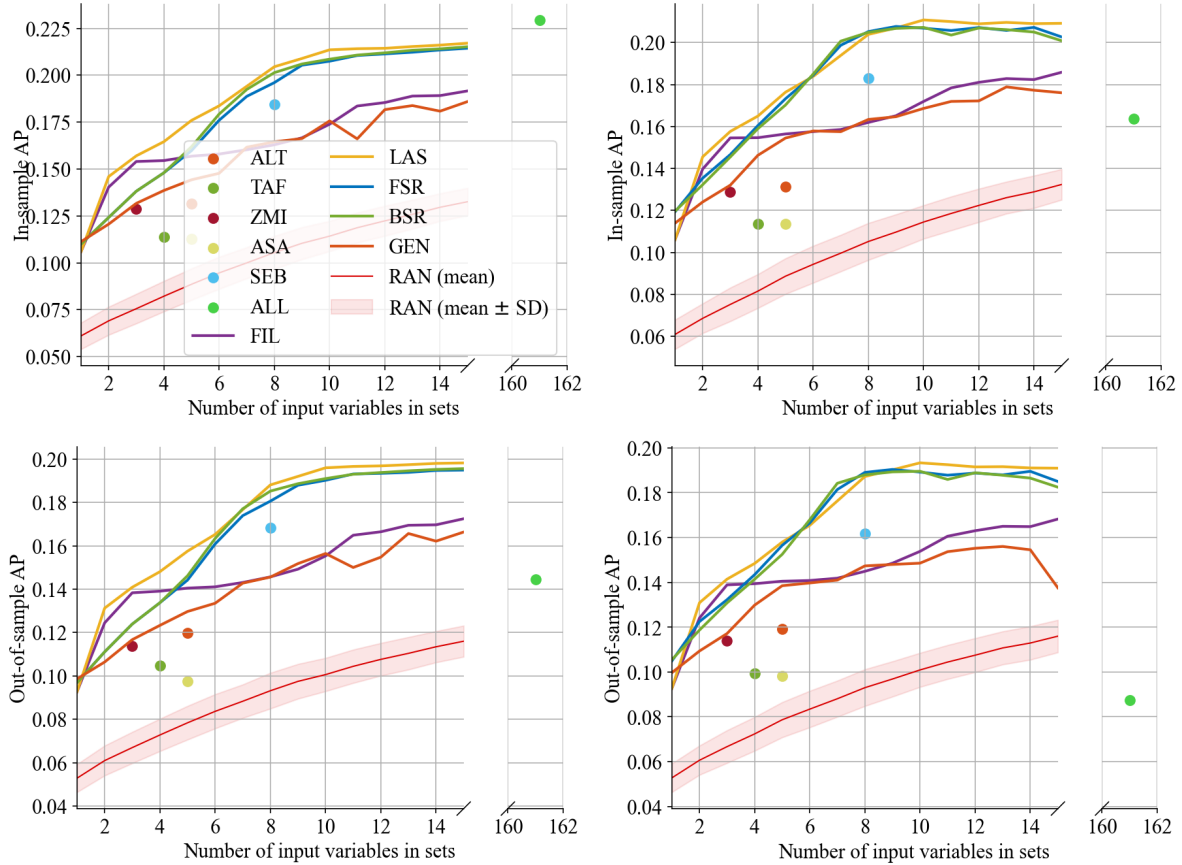
Figures IA.A3 and IA.A4 reconstruct Figure 2 of the manuscript (which evaluates models using AP), but for predicting bankruptcy two and three years ahead, respectively. Similarly, Figures IA.A5 and IA.A6 replicate Figure IA.1 (which evaluates models using AUC) for predicting bankruptcies two and three years ahead. Furthermore, Figure IA.A7 replicates Figure 3 of the manuscript, demonstrating the stability of variable selection for two- and three-year-ahead prediction horizons. These results show that our main findings are robust to predicting bankruptcy over horizons extending one year.

Figure IA.A3: AP obtained when predicting two-year-ahead bankruptcy



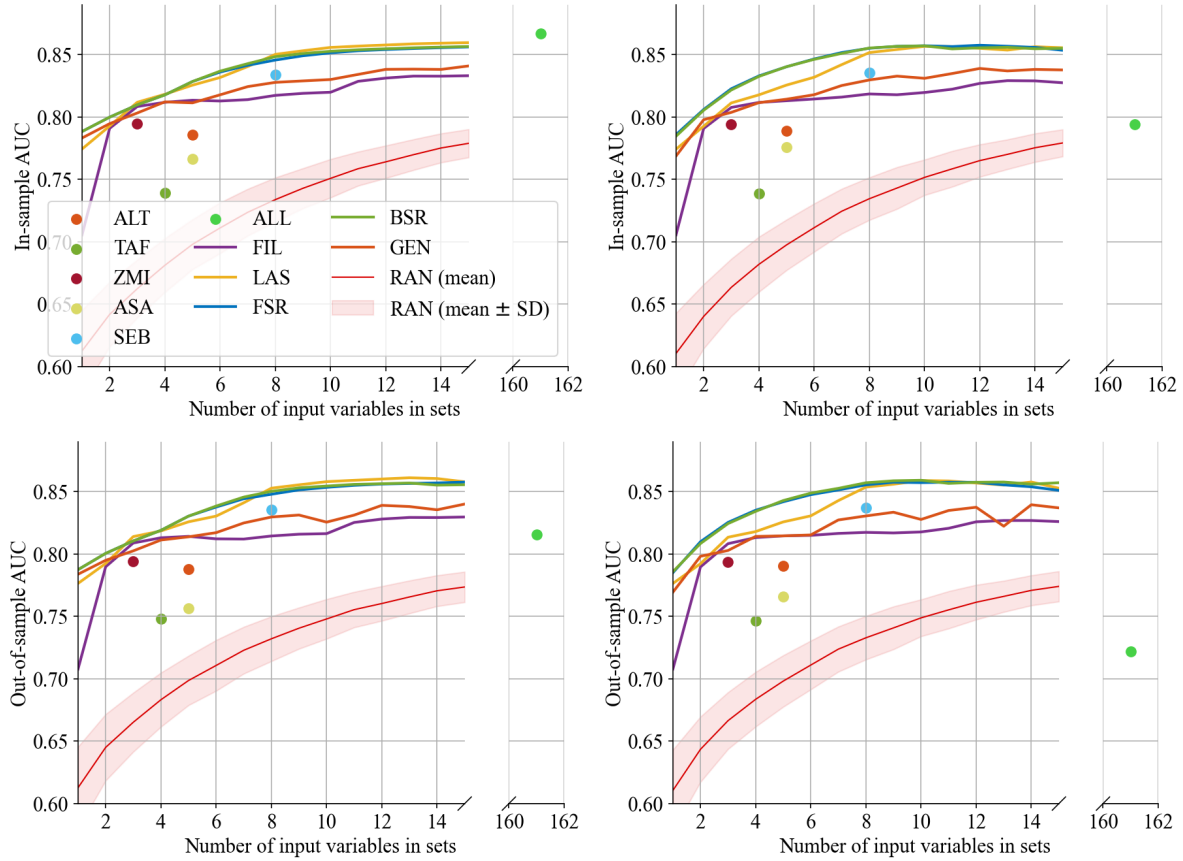
Financial statements are categorized as bankrupt if they are the last or second last financial statement of a company that is filed for bankruptcy within three years from the balance sheet date of its last financial statement. The AP values are the mean of AP across the eight evaluations for the years 2010-2020. For FSR, BSR, GEN, LAS, FIL, and RAN, we let the predetermined number of input variables in the sets vary between 1 and 15. Moreover, for RAN, we report the mean value and the standard deviation range (mean $\pm$ one standard deviation). Top and bottom panels show in-sample AP and out-of-sample AP, respectively. Left and right panels display results when using LR and DNN, respectively, as estimation technique.

Figure IA.A4: AP obtained when predicting three-year-ahead bankruptcy



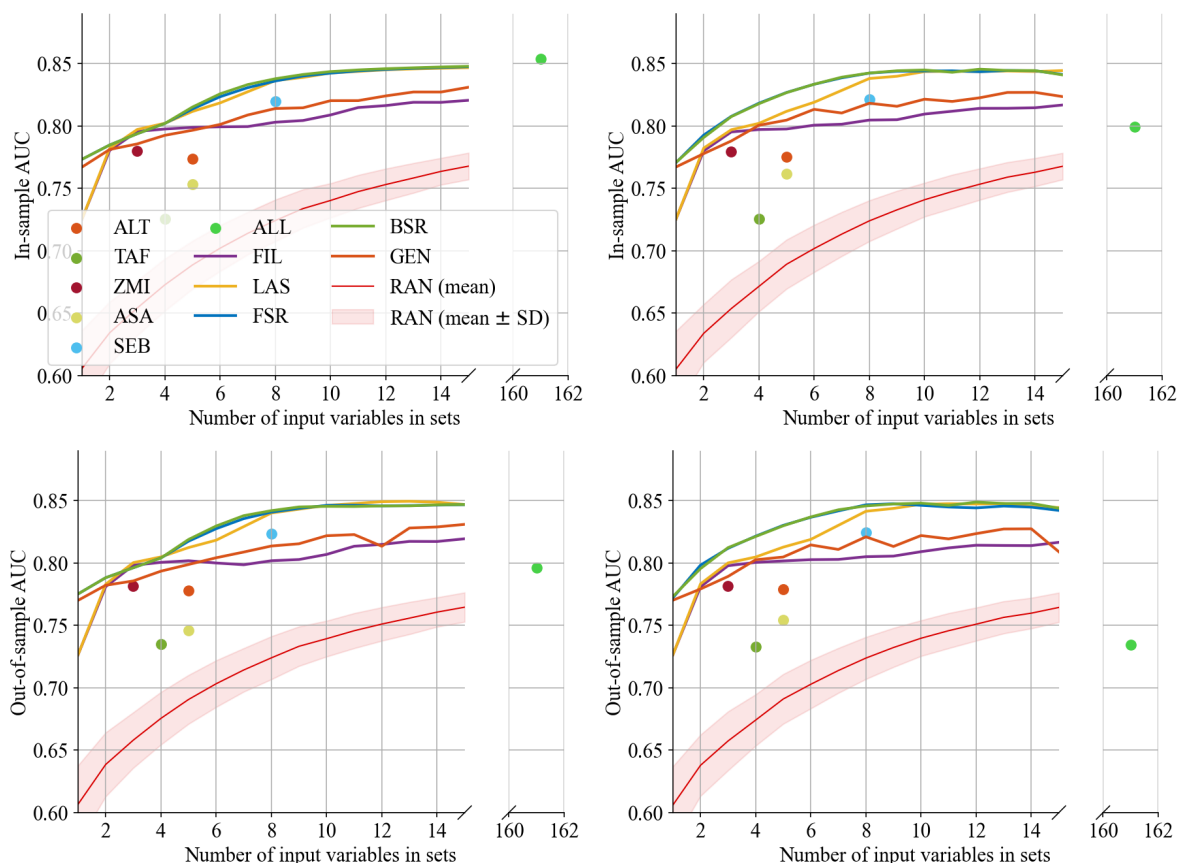
Financial statements are categorized as bankrupt if they are the last, second last, or third last financial statement of a company that is filed for bankruptcy within three years from the balance sheet date of its last financial statement. The AP values are the mean of AP across the eight evaluations for the years 2010-2020. For FSR, BSR, GEN, LAS, FIL, and RAN, we let the predetermined number of input variables in the sets vary between 1 and 15. Moreover, for RAN, we report the mean value and the standard deviation range (mean $\pm$ one standard deviation). Top and bottom panels show in-sample AP and out-of-sample AP, respectively. Left and right panels display results when using LR and DNN, respectively, as estimation technique.

Figure IA.A5: AUC obtained when predicting two-year-ahead bankruptcy



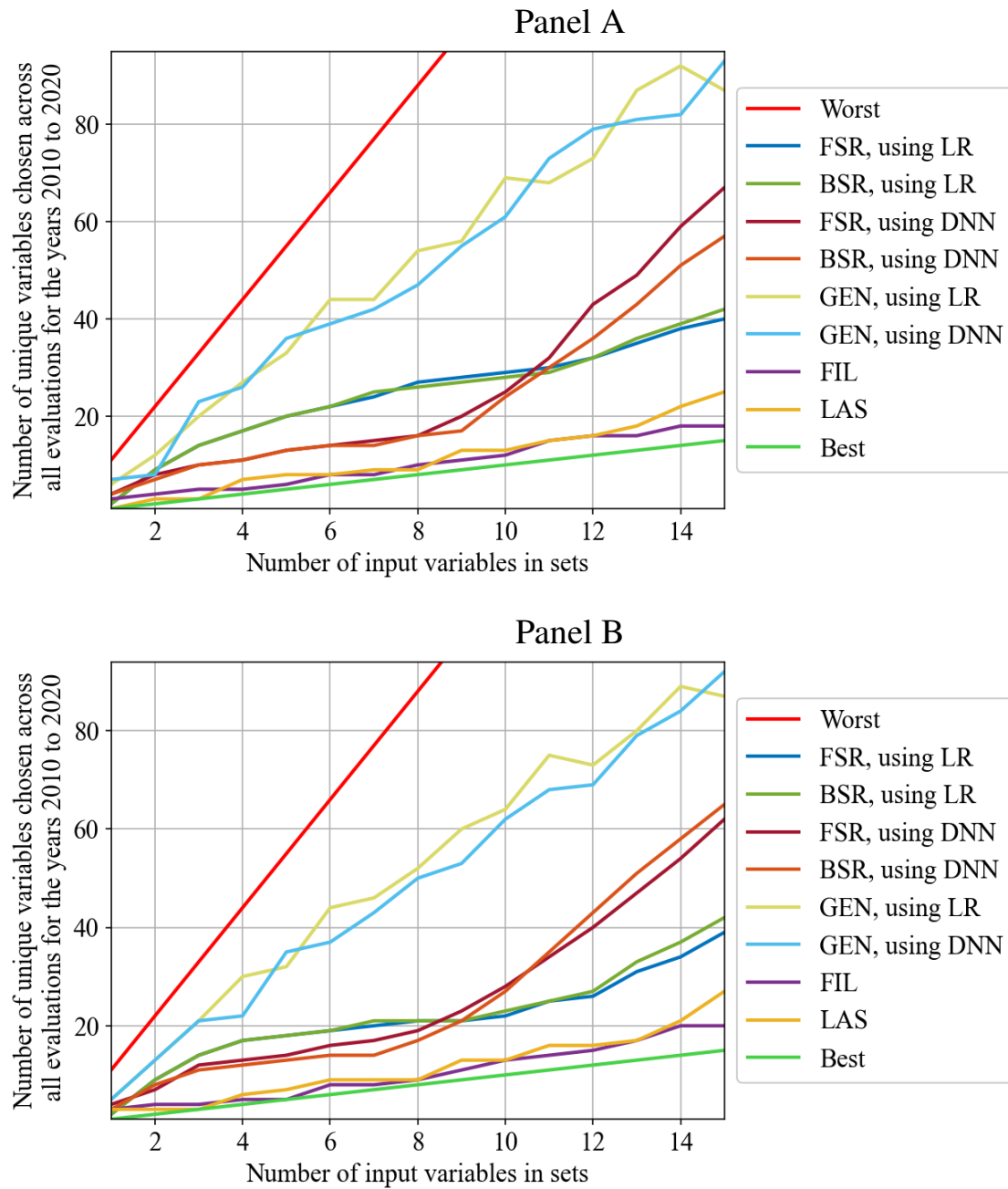
Financial statements are categorized as bankrupt if they are the last or second last financial statement of a company that is filed for bankruptcy within three years from the balance sheet date of its last financial statement. The AUC values are the mean of AUC across the eight evaluations for the years 2010-2020. For FSR, BSR, GEN, LAS, FIL, and RAN, we let the predetermined number of input variables in the sets vary between 1 and 15. Moreover, for RAN, we report the mean value and the standard deviation range (mean $\pm$ one standard deviation). Top and bottom panels show in-sample AUC and out-of-sample AUC, respectively. Left and right panels display results when using LR and DNN, respectively, as estimation technique.

Figure IA.A6: AUC obtained when predicting three-year-ahead bankruptcy



Financial statements are categorized as bankrupt if they are the last, second last, or third last financial statement of a company that is filed for bankruptcy within three years from the balance sheet date of its last financial statement. The AUC values are the mean of AUC across the eight evaluations for the years 2010-2020. For FSR, BSR, GEN, LAS, FIL, and RAN, we let the predetermined number of input variables in the sets vary between 1 and 15. Moreover, for RAN, we report the mean value and the standard deviation range (mean $\pm$ one standard deviation). Top and bottom panels show in-sample AUC and out-of-sample AUC, respectively. Left and right panels display results when using LR and DNN, respectively, as estimation technique.

Figure IA.A7: Stability of variable selection when predicting bankruptcy two and three years ahead



Panels A and B show the total number of unique input variables chosen by each feature selection method when predicting bankruptcy two and three years ahead, respectively. These variables are chosen across the 11 evaluations over the years 2010-2020 for different predetermined number of variables in sets. The green and red diagonal lines represent the best and worst possible cases, respectively.

Appendix IA.4: Technical details of the simulation study

In the simulation study, the competitive credit market is structured as follows: Let $\mathbf{r} = \{r_n\}_{n=1,\dots,N}$ be the vector of credit spreads assigned by one of the banks with n denoting the potential borrowers (financial statements). We assume that all banks set the credit spreads such that the expected revenues cover all expected losses plus profit for each loan:

$$\underbrace{(1 - \hat{y}_n) \times (r_n + c)}_{\text{conditional expected revenue}} = \underbrace{\hat{y}_n \times (LGD + r_f)}_{\text{conditional expected loss}} + \underbrace{(1 - \hat{y}_n) \times k}_{\text{conditional expected profit}} \quad (\text{IA.6})$$

where $\hat{y}_n$ is the predicted out-of-sample probability for bankruptcy derived from the bank's specific model, LGD is the loss-given default in the case of borrower bankruptcy (including recovery costs), r_f is the risk-free rate, k is the credit spread for a loan with no credit risk, i.e., the required profit margin, and c is the cost of granting loans for strategic reasons.¹ Similar to Blöchlinger and Leippold (2006), Agarwal and Taffler (2008), and Bauer and Agarwal (2014), we assume $c = 0$. Consequently, equation (IA.6) can be rewritten to the following equation for deriving the credit spreads assigned to each potential borrower:

$$r_n = \frac{\hat{y}_n}{1 - \hat{y}_n} (LGD + r_f) + k \quad (\text{IA.7})$$

We follow the criteria of rejection based on the credit spread values as in Blöchlinger and Leippold (2006), and assume that banks reject potential borrowers when $r_n > 10\%$. Further, we assume $LGD = 40\%$, which is in line with the Basel III regulatory framework (Bank of International Settlements, 2017). In addition, we assume $k = 0.30\%$, following Blöchlinger and Leippold (2006), Agarwal and Taffler (2008), and Bauer and Agarwal (2014).

If the potential borrowers are granted loans from several banks, we assume they choose to borrow the whole amount from the bank offering the lowest credit spread. Further, we follow the

¹If the bank expects profits from a customer relationship through businesses other than lending, it may be willing to lower credit spreads in order to retain or acquire the customer. The cost for such lowering of credit spreads are captured by c .

literature and set no limits on the total amount any bank can lend. In addition, if borrowers go bankrupt, we assume that the lending bank receives no interest payment and suffers a loss equal to the principal weighted by LGD. In sum, the total profit of any bank is given by

$$\underbrace{\sum_{n=1}^N p_n \times r_n \times (1 - y_n)}_{\text{total revenue}} - \underbrace{\sum_{n=1}^N p_n \times (LGD + r_f) \times y_n}_{\text{total loss}} \quad (\text{IA.8})$$

where p_n is the principal, i.e., the interest bearing debt derived from the financial statement, and y_n is the actual classification of bankrupt (1) or non-bankrupt (0). The first term of equation (IA.8) represents the bank's total revenue. The bank's revenue increases when the bank lends to more non-bankrupting borrowers as a result of using a variable set that yields lower credit spreads for these non-bankrupting customers compared to competing banks. Hence, avoiding the second type of error, as explained above, results in higher revenues. The second term of equation (IA.8) measures the bank's total loss. The bank's loss declines when a bank uses a variable set that rejects loans to bankrupting potential borrowers by providing more accurate predictions about their bankruptcies, i.e., by avoiding the first type of error explained above. In summary, equation (IA.8) measures a bank's total profit that results from a given variable set when taking into account realistic costs brought about by the first and second type of prediction errors.

Appendix IA.5: Selecting variable sets at the industry-level

We evaluate the choice of bankruptcy predictors at the industry level by selecting industry-specific variable sets for all industries that include at least 500 financial statements that are categorized as bankrupt within the entire sample period. To keep the analysis traceable, we rely on our main setting, i.e., we use LASSO in combination with discrete hazard models estimated with LR, and restrict the selection of bankruptcy predictors to nine variables for each of the years 2010-2020.

Table IA.A1, panels A to H, shows the bankruptcy predictors chosen for each industry and year. Each of the eight panels summarizes results for a specific industry.² Results show that the variables chosen in the full sample, as reported in Table 4, are quite persistent across the 88 permutations, resulting from 11 yearly analyses across eight different industries: In eight permutations, the variable set selected at the industry-level exactly matches that selected in the full sample. In 27 permutations, one of nine bankruptcy predictors is replaced. In 41 permutations, two predictors are replaced. In only twelve permutations, three or four predictors are replaced.

The key question is, of course, whether selecting bankruptcy predictors at the industry-level improves model performance in a way that matters to bank profitability. To test this empirically, we run a simulation exercise with two banks similar to that in Section 5 of the manuscript. The first bank uses LAS, which is the most profitable approach from the simulation described in Section 5, with results reported in Table 5 of the manuscript. That is, the bank selects variables with the LASSO method applied to the full sample. The second bank uses the same bankruptcy prediction model, but selects variables for each industry separately and applies them to potential borrowers according to their industry affiliation (referred to as LIN). For borrowers in industries that do not qualify for a separate bankruptcy prediction model estimation due a small number of bankrupt ob-

²We select industry-specific bankruptcy predictors for the industries 'Manufacturing', 'Construction', 'Wholesale and retail trade; repair of motor vehicles and motorcycles', 'Transportation and storage', 'Accommodation and food service activities', 'Information and communication', 'Professional, scientific and technical activities', and 'Administrative and support service activities'. We omit the following industries from this analysis because of a small number of financial statements that are categorized as bankrupt: 'Agriculture, forestry and fishing', 'Mining and quarrying', 'Education', 'Human health and social work activities', 'Arts, entertainment and recreation', and 'Other service activities'. Note that there are no firms in industries 'Activities of household as employers' or 'Activities of extraterritorial organisations and bodies', which are part of the Norwegian SIC2007, in our sample.

servations as explained above, we use the model configured to the full sample (i.e., LAS). Results obtained from simulating the credit market with these two banks are reported in Table IA.A2.³ The table shows that LIN is associated with a profit margin that is higher by 15%, 1% higher ROA, and 21% higher RORWA.

We conduct a number of robustness tests for these results. For instance, we find that the superiority of LIN in terms of bank profitability is robust to defining the amount to be borrowed by each company either in terms of interest-bearing debt or total debt. Specifically, in both cases, LIN is superior across all combinations of loss-given default (LGD) $\in \{30\%, 40\%, 50\%, 60\%, 70\%\}$ and required profit margin (k) $\in \{0.1\%, 0.2\%, 0.3\%, 0.4\%, 0.5\%\}$.

In summary, the results in this section show that the flexibility of our feature selection-based bankruptcy prediction models allows to not only reconfigure models over time, but also to estimate models across sub-samples, such as for instance across industries, to further improve model performance. Results show that while our model is fairly stable across both time and industries, estimating our model at the industry-level further improves model performance and eventually the profitability of banks applying these models.

³As in Table 5 of the manuscript, neither the total market share nor the share of bankrupting borrowers in the market add up to 100% as several potential borrowers are rejected by both banks.

Table IA.A1: Industry-specific variable sets

Panel A: Manufacturing

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Variables selected when using all data	dummy; one if total liability exceeds total assets	X	X	X	X	X	X	X	X	X	X
	accounts payable / total assets	X	X	X	X	X	X	X	X	X	X
	(current liabilities - short-term liquidity) / total assets	X	X	X	X	X	X	X	X	X	X
	net income / total assets	X	X	X	X	X	X	X	X	X	X
	public taxes payable / total assets	X	X	X	X	X	X	X	X	X	X
	interest expenses / total assets	X	X	X	X	X	X	X	X	X	X
	dummy; one if paid-in equity is less than total equity	X	X	X	X	X	X	X	X	X	X
	log(age in years)										
	short-term liquidity / current assets						X				
Variables not selected when using all data	total equity / total assets	X	X	X	X	X				X	X
	inventory / current assets	X									
	total liabilities / total assets		X		X	X					
	pre-tax profit / total assets						X	X	X		
	intangible assets / total assets							X			
	EBIT / tangible assets								X	X	X

The nine input variables selected for the years 2010–2020 by the LASSO method when applied to financial statements of companies from the industry 'Manufacturing'. The upper part of the table includes variables also chosen when using the full sample (see Table 4). The lower part of the table lists other variables that are chosen in this specific industry for the respective year. The industry includes 141,118 observations, 1,946 of which are classified as bankrupt (1.38%).

Panel B: Construction

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Variables selected when using all data	dummy; one if total liability exceeds total assets	X	X	X	X	X	X	X	X	X	X
	accounts payable / total assets	X	X	X	X	X	X	X	X	X	X
	(current liabilities - short-term liquidity) / total assets	X	X	X	X	X	X	X	X	X	X
	net income / total assets	X	X	X	X	X	X	X	X	X	X
	public taxes payable / total assets	X	X	X	X	X	X	X	X	X	X
	interest expenses / total assets		X	X	X	X	X	X	X		
	dummy; one if paid-in equity is less than total equity	X	X	X	X	X					
	log(age in years)	X	X	X	X	X	X	X	X	X	X
	short-term liquidity / current assets			X	X	X	X	X	X	X	X
Variables not selected when using all data	total equity / total assets	X	X	X							
	retained earnings / total assets	X									
	sales / capital employed						X				
	current liabilities / total assets							X	X	X	X
	cost of goods sold / inventory								X	X	X

The nine input variables selected for the years 2010-2020 by the LASSO method when applied to financial statements of companies from the industry 'Construction'. The upper part of the table includes variables also chosen when using the full sample (see Table 4). The lower part of the table lists other variables that are chosen in this specific industry for the respective year. The industry includes 374,177 observations, 6,188 of which are classified as bankrupt (1.65%).

Panel C: Wholesale and retail trade; repair of motor vehicles and motorcycles

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Variables selected when using all data											
dummy; one if total liability exceeds total assets	X	X	X	X	X	X	X	X	X	X	X
accounts payable / total assets	X	X	X	X	X	X	X	X	X	X	X
(current liabilities - short-term liquidity) / total assets	X	X	X	X	X	X	X	X	X	X	X
net income / total assets	X	X	X	X	X	X	X	X	X	X	X
public taxes payable / total assets											
interest expenses / total assets	X	X	X	X	X	X	X	X	X	X	X
dummy; one if paid-in equity is less than total equity	X	X	X	X	X	X	X	X	X	X	X
log(age in years)											
short-term liquidity / current assets		X	X	X	X	X	X	X	X	X	X
Variables not selected when using all data											
retained earnings / total assets	X										
total equity / total assets	X	X	X	X	X					X	X
pre-tax profit / total assets	X										
inventory / current assets		X	X	X	X	X	X	X			
log(total assets)									X		

The nine input variables selected for the years 2010-2020 by the LASSO method when applied to financial statements of companies from the industry 'Wholesale and retail trade; repair of motor vehicles and motorcycles'. The upper part of the table includes variables also chosen when using the full sample (see Table 4). The lower part of the table lists other variables that are chosen in this specific industry for the respective year. The industry includes 448,781 observations, 8,264 of which are classified as bankrupt (1.84%).

Panel D: Transportation and storage

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Variables selected when using all data											
dummy; one if total liability exceeds total assets	X	X	X	X	X	X	X	X	X	X	X
accounts payable / total assets	X	X	X	X	X	X	X	X	X	X	X
(current liabilities - short-term liquidity) / total assets	X	X	X	X	X	X	X	X	X	X	X
net income / total assets				X	X	X	X	X	X	X	X
public taxes payable / total assets	X	X	X	X	X	X	X	X	X	X	X
interest expenses / total assets	X	X	X	X	X	X	X	X	X	X	X
dummy; one if paid-in equity is less than total equity	X	X	X	X							
log(age in years)	X	X	X	X	X	X	X	X	X	X	X
short-term liquidity / current assets	X	X	X	X	X	X	X	X			
Variables not selected when using all data											
operating profit / total assets	X	X	X								
sales / current assets				X						X	X
total equity / (total equity + long-term liabilities)						X	X		X	X	X
log(total assets)								X	X	X	X

The nine input variables selected for the years 2010-2020 by the LASSO method when applied to financial statements of companies from the industry 'Transportation and storage'. The upper part of the table includes variables also chosen when using the full sample (see Table 4). The lower part of the table lists other variables that are chosen in this specific industry for the respective year. The industry includes 100,138 observations, 1,305 of which are classified as bankrupt (1.30%)

Panel E: Accommodation and food service activities

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Variables selected when using all data	dummy; one if total liability exceeds total assets	X	X	X	X	X	X	X	X	X	X
	accounts payable / total assets	X	X	X	X	X	X	X	X	X	X
	(current liabilities - short-term liquidity) / total assets				X	X	X	X	X	X	
	net income / total assets	X	X	X	X	X	X	X	X	X	X
	public taxes payable / total assets	X	X	X	X	X	X	X	X	X	X
	interest expenses / total assets	X	X	X	X	X	X	X			
	dummy; one if paid-in equity is less than total equity	X	X		X		X	X	X	X	X
	log(age in years)	X	X	X	X	X	X	X	X	X	X
	short-term liquidity / current assets								X		
Variables not selected when using all data	inventory / current assets	X	X	X	X						
	log(total assets)	X	X				X	X	X	X	X
	total equity / total assets			X	X	X					
	working capital / total assets			X				X			
	sales / current assets			X							
	operating profit / total assets					X	X			X	X
	square of change in 10-year synthetic bond yields										X

The nine input variables selected for the years 2010-2020 by the LASSO method when applied to financial statements of companies from the industry 'Accommodation and food service activities'. The upper part of the table includes variables also chosen when using the full sample (see Table 4). The lower part of the table lists other variables that are chosen in this specific industry for the respective year. The industry includes 81,543 observations, 2,151 of which are classified as bankrupt (2.64%).

Panel F: Information and communication

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Variables selected when using all data											
dummy; one if total liability exceeds total assets	X	X	X	X	X	X	X	X	X	X	X
accounts payable / total assets	X	X	X	X	X	X	X	X	X	X	X
(current liabilities - short-term liquidity) / total assets	X	X	X	X	X	X	X	X	X	X	X
net income / total assets	X	X	X	X	X	X					
public taxes payable / total assets	X	X	X	X	X	X	X	X	X	X	X
interest expenses / total assets	X	X	X	X	X	X	X	X	X	X	X
dummy; one if paid-in equity is less than total equity	X	X	X	X	X	X	X	X	X	X	X
log(age in years)		X			X		X				
short-term liquidity / current assets		X		X	X						
Variables not selected when using all data											
pre-tax profit / total assets	X										
operating profit / total assets	X			X				X	X	X	X
total equity / total assets			X								
short-term liquidity / total assets			X								
log(total assets)				X							
EBIT / tangible assets						X	X	X	X	X	X
sales / current assets						X					
(current assets - total liabilities) / total assets							X	X			
total liabilities / total assets									X	X	X

The nine input variables selected for the years 2010-2020 by the LASSO method when applied to financial statements of companies from the industry '*Information and communication*'. The upper part of the table includes variables also chosen when using the full sample (see Table 4). The lower part of the table lists other variables that are chosen in this specific industry for the respective year. The industry includes 99,635 observations, 757 of which are classified as bankrupt (0.76%).

Panel G: Professional, scientific and technical activities

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Variables selected when using all data	dummy; one if total liability exceeds total assets	X	X	X	X	X	X	X	X	X	X
	accounts payable / total assets	X	X	X	X	X	X	X	X	X	X
	(current liabilities - short-term liquidity) / total assets	X	X	X	X	X	X	X	X	X	X
	net income / total assets	X	X	X	X	X	X	X			
	public taxes payable / total assets	X	X	X	X	X	X		X	X	X
	interest expenses / total assets	X	X	X	X	X	X		X	X	X
	dummy; one if paid-in equity is less than total equity	X	X	X	X	X	X	X	X	X	X
	log(age in years)										
	short-term liquidity / current assets		X								
Variables not selected when using all data	retained earnings / total assets	X		X			X	X	X		
	pre-tax profit / total assets	X									
	accounts payable / current liabilities		X			X					
	intangible assets / total assets					X					
	(non-interest expenses - labour costs) / total assets				X			X	X	X	X
	total expenses / total assets						X	X			
	sales / current assets							X			X
	operating expenses / total assets								X		
	total equity / total assets										X

The nine input variables selected for the years 2010-2020 by the LASSO method when applied to financial statements of companies from the industry 'Professional, scientific and technical activities'. The upper part of the table includes variables also chosen when using the full sample (see Table 4). The lower part of the table lists other variables that are chosen in this specific industry for the respective year. The industry includes 270,205 observations, 1,466 of which are classified as bankrupt (0.54%).

Panel H: Administrative and support service activities

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Variables selected when using all data											
dummy; one if total liability exceeds total assets	X	X	X	X	X	X	X	X	X	X	X
accounts payable / total assets	X	X	X	X	X	X	X	X	X	X	X
(current liabilities - short-term liquidity) / total assets	X	X	X	X	X	X	X	X	X	X	X
net income / total assets	X	X	X	X	X	X	X	X	X	X	X
public taxes payable / total assets	X	X	X	X	X	X	X	X	X	X	X
interest expenses / total assets	X	X	X	X	X	X	X	X	X	X	X
dummy; one if paid-in equity is less than total equity	X	X	X	X	X	X	X	X	X	X	X
log(age in years)	X	X	X	X	X	X	X	X	X	X	X
short-term liquidity / current assets	X	X	X	X	X	X	X	X	X	X	X
Variables not selected when using all data											
cost of goods sold / inventory	X	X	X	X	X	X	X	X	X	X	X
working capital / total equity					X					X	X
total equity / total assets							X		X	X	X
sales / current assets											
log(total assets)								X	X	X	X

The nine input variables selected for the years 2010-2020 by the LASSO method when applied to financial statements of companies from the industry 'Administrative and support service activities'. The upper part of the table includes variables also chosen when using the full sample (see Table 4). The lower part of the table lists other variables that are chosen in this specific industry for the respective year. The industry includes 92,272 observations, 1,218 of which are classified as bankrupt (1.32%).

Table IA.A2: Simulation of a competitive credit market with two banks – general versus industry-specific datasets

	LAS	LIN
Market share of each bank in EUR (%)	26.4	73.6
Share of bankrupting borrowers in portfolio (%)	1.5	1.0
Share of bankrupting borrowers in the market (%)	35.5	26.6
Revenue	3,640.6	8,853.0
Loss	925.7	1,231.0
Profit	2,714.9	7,622.0
Profit margin (%)	74.6	86.1
Return on assets (ROA) (%)	0.4	0.4
Return on risk-weighted assets (RORWA) (%)	1.0	1.2

The first bank in the simulation uses the variable set chosen by the LASSO method when using the full sample (LAS). The second bank, LIN, uses for each potential borrower the variable set chosen by LASSO that corresponds to the potential borrower's industry (see Table IA.A1, panels A to H) or the variable set chosen by the overall model, as in LAS, for industries that are too small to select a separate variable set. The table shows the market share for each bank, as well as the share of borrowers that went bankrupt in each bank's loan portfolio and the share of all borrowers in the market that went bankrupt who borrow from each bank. Further, it reports revenues, losses, and profits of each bank in EUR millions. Finally, the table presents profit margins, return on assets (ROA), which is profit divided by total amount lent, and return on risk-weighted assets (RORWA), calculated by following the Basel III regulatory framework. We use in our simulation all financial statements in our sample from the accounting years 2010-2020 (see Figure 1 of the manuscript).

Table IA.1: Input variables for bankruptcy prediction

The table lists all 161 input variables considered in our study. It also reports the Pearson correlation coefficient (PCC) between the values of the input variables and the actual classifications of non-bankrupt (0) or bankrupt (1) for each financial statement. The sample consists of 1,822,784 financial statements whereas 1.35% are categorized as bankrupt. Input variables are sorted according to the absolute value of PCC (in descending order).

Description	PCC
dummy; one if total liability exceeds total assets	0.15
accounts payable / total assets	0.14
total equity / total assets	-0.14
total liabilities / total assets	0.14
(current liabilities - short-term liquidity) / total assets	0.14
current liabilities / total assets	0.13
retained earnings / total assets	-0.13
net income / total assets	-0.12
dummy; one if paid-in equity is less than total equity	-0.12
pre-tax profit / total assets	-0.12
EBIT / total assets	-0.11
working capital / total assets	-0.11
EBIT / tangible assets	-0.11
operating profit / total assets	-0.11
interest expenses / total assets	0.11
EBITDA / total assets	-0.11
(current assets - total liabilities) / total assets	-0.10
public taxes payable / total assets	0.10
total expenses / total assets	0.08
operating expenses / total assets	0.08
(non-interest expenses - labour costs) / total assets	0.07
sales / current assets	0.07
sales / tangible assets	0.07
accounts payable / current liabilities	0.07
(paid-in capital + total revenues) / total assets	0.07
sales / total assets	0.07
short-term liquidity / current assets	-0.07
investment turnover (sales / (total equity + total liabilities))	0.07
total revenues / total assets	0.07
log(age in years)	-0.07
labour costs / total assets	0.06
(current assets - short-term liquidity) / total assets	0.06
log(total assets)	-0.06
net income / (total liabilities + paid-in capital)	-0.06
short-term liquidity / total assets	-0.05
inventory / current assets	0.05
effective tax rate	-0.05

Table IA.1: Input variables in our study, descending by the absolute value of PCC (continued)

Description	PCC
total equity / (total equity + long-term liabilities)	-0.03
dividends / net income	-0.03
interest income / total assets	-0.03
operating expenses / sales	-0.02
short-term liquidity / capital employed	-0.02
interest income / sales	-0.02
long-term liabilities / total assets	0.02
short-term liquidity / sales	-0.02
interest income / interest expenses	-0.02
(total revenues - sales) / total revenues	-0.02
net income / total revenues	-0.02
operating profit / sales	-0.02
intangible assets / total assets	0.02
total assets / total revenues	-0.02
working capital / sales	-0.02
sales / working capital	-0.02
retained earnings / inventory	-0.02
quick assets / total assets	-0.02
working capital / total revenues	-0.02
total revenues / working capital	-0.02
sales / intangible assets	0.02
sales / total equity	-0.02
labour costs / operating profit	-0.02
total liabilities / total equity	-0.02
fixed assets / total equity	-0.02
current liabilities / total equity	-0.02
EBITDA / interest expense	-0.01
current assets / total equity	-0.01
fixed assets / total assets	-0.01
current assets / total assets	0.01
EBIT / interest expenses	-0.01
long-term liabilities / total equity	-0.01
interest-bearing debt / total equity	-0.01
total revenues / sales	-0.01
net income / operating profit	0.01
current assets / sales	-0.01
interest expenses / total revenues	-0.01
(quick_assets - current_liabilities) / inventory	-0.01
retained earnings / sales	-0.01
quick assets / sales	-0.01
operating profit / total revenues	-0.01
income gearing	-0.01

Table IA.1: Input variables in our study, descending by the absolute value of PCC (continued)

Description	PCC
current liabilities / sales	-0.01
change in unemployment rate	-0.01
non-interest expenses / operating profit	-0.01
change in 10-year synthetic bond yields	0.01
working capital / operating expenses	-0.01
accounts payable / sales	-0.01
total equity / sales	-0.01
accounts receivable / accounts payable	-0.01
net income / sales	-0.01
pre-tax profit / sales	-0.01
EBIT / sales	-0.01
cost of goods sold / inventory	0.01
inventory / working capital	-0.01
net income / total equity	0.01
accounts receivable / sales	-0.01
operating profit / (operating profit - interest expenses)	-0.01
quick assets / current liabilities	-0.01
pre-tax profit / ordinary expenses	-0.01
working capital / current liabilities	-0.01
paid-in capital / total assets	0.01
financial expenses / sales	-0.01
share of labour costs	0.01
current liabilities / total liabilities	0.01
(quick assets / current liabilities) X interest earned ratio	-0.01
intangible assets / sales	-0.01
(total equity - intangible assets) / (tangible assets - short-term liquidity)	-0.01
total equity / long-term liabilities	-0.01
short-term liquidity / current liabilities	-0.01
(inventory + accounts receivable) / total equity	-0.01
accounts payable / inventory	-0.01
retained earnings / current liabilities	-0.01
retained earnings / intangible assets	0.01
current liabilities / EBIT	-0.01
interest expenses / total expenses	-0.01
inventory / cost of goods sold	-0.01
pre-tax profit / current liabilities	-0.01
long-term liabilities / current assets	-0.01
pre-tax profit / total equity	0.01
return on debt (net income / total liabilities)	-0.01
(net income + depreciations + amortizations) / total liabilities	-0.01
EBIT / total liabilities	-0.00
EBITDA / total liabilities	-0.00

Table IA.1: Input variables in our study, descending by the absolute value of PCC (continued)

Description	PCC
pre-tax profit / capital employed	-0.00
current assets / total liabilities	-0.00
working capital / long-term liabilities	-0.00
total equity / total liabilities	-0.00
operation asset / total assets	0.00
sales / paid-in capital	0.00
change in CPI growth rate	0.00
total revenues / total expenses	-0.00
(sales - cost of goods sold) / sales	-0.00
interest bearing debt / total liabilities	-0.00
sales / accounts receivable	-0.00
inventory / current liability	-0.00
(total revenues + interest income) / total expenses	-0.00
cost of goods sold / sales	-0.00
short-term liquidity / total liabilities	-0.00
total equity / fixed assets	-0.00
sales / capital employed	-0.00
working capital / total equity	0.00
net income / capital employed	-0.00
accounts receivable / total liabilities	-0.00
operating profit / capital employed	-0.00
(long-term liabilities + total equity) / fixed assets	-0.00
inventory / sales	-0.00
change in GDP growth rate	0.00
net income / paid-in capital	0.00
pre-tax profit / paid-in capital	0.00
no-credit interval	-0.00
current assets / paid-in capital	0.00
sales / inventory	-0.00
fixed assets / (paid-in capital + long-term liabilities)	0.00
sales / short-term liquidity	-0.00
total revenues / fixed assets	-0.00
operating profit / paid-in capital	-0.00
square of change in 10-year synthetic bond yields	-0.00
return on fixed assets	0.00
change in term spreads	0.00
interest expenses / total liabilities	0.00
current liabilities / current assets	-0.00
sales / fixed assets	-0.00
profits / working capital	0.00

Table IA.2: Input variables in the benchmark sets

The model of Altman (1968) (ALT)

EBIT / total assets
retained earnings / total assets
sales / total assets
total equity / total liabilities
working capital / total assets

The model of Taffler (1984) (TAF)

current assets / total liabilities
current liabilities / total assets
no-credit interval
pre-tax profit / current liabilities

The model of Zmijewski (1984) (ZMI)

current liabilities / current assets
net income / total assets
total liabilities / total assets

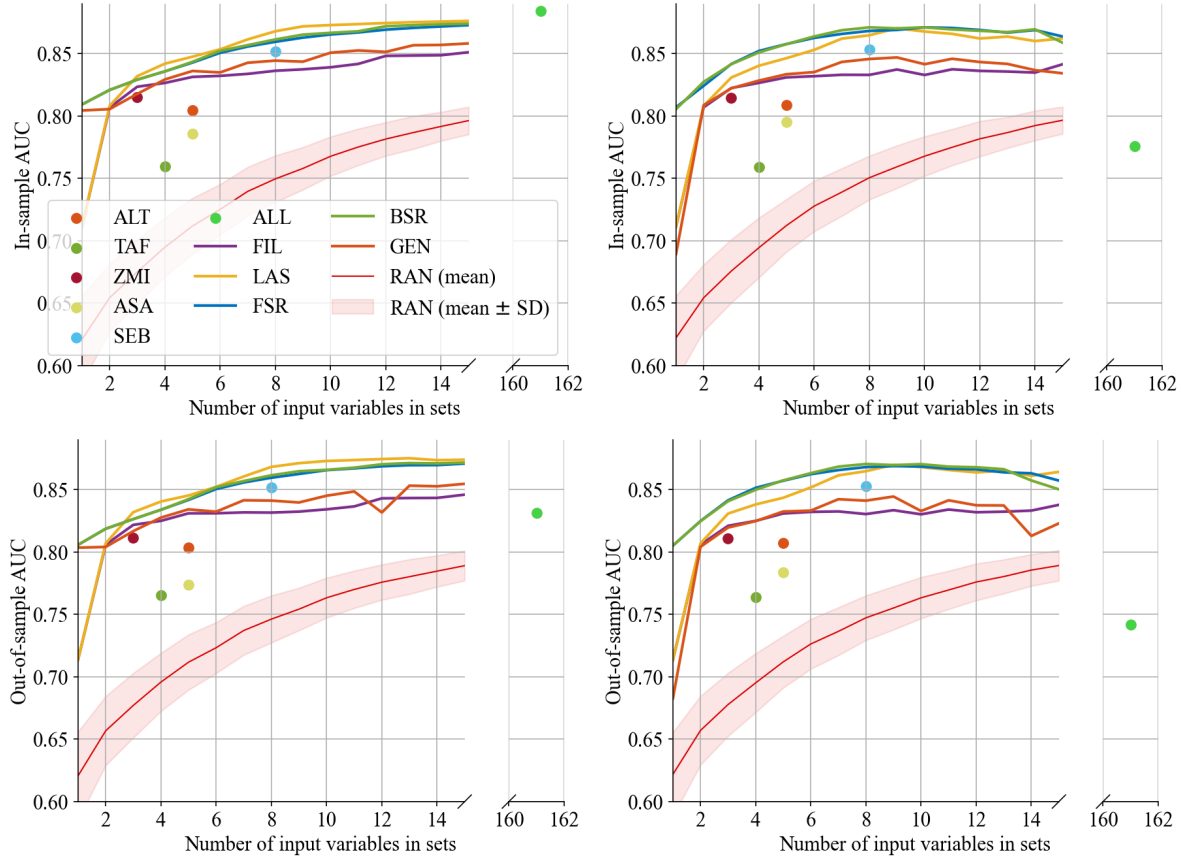
The model of Altman and Sabato (2007) (ASA)

current liabilities / total equity
EBITDA / interest expense
EBITDA / total assets
retained earnings / total assets
short-term liquidity / total assets

The SEBRA model (Bernhardsen and Larsen, 2007) (SEB)

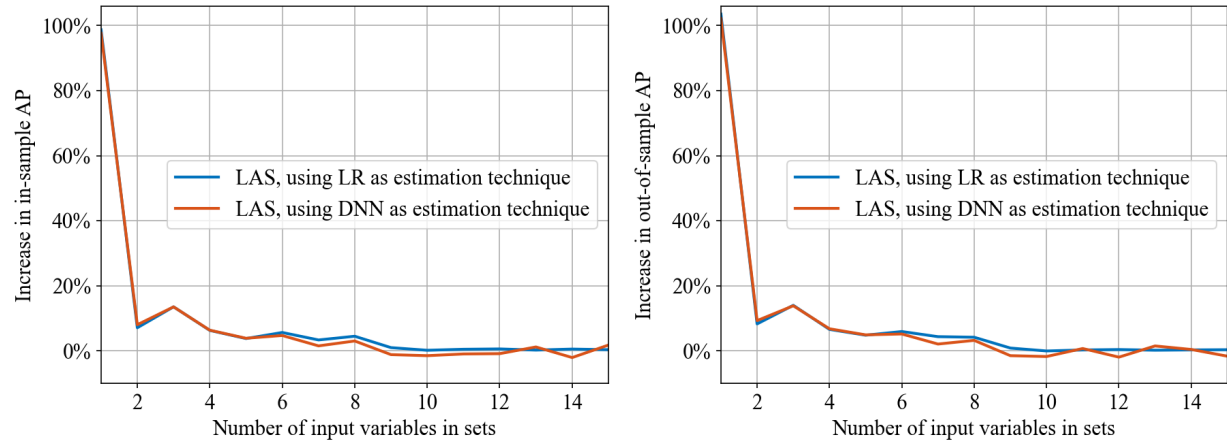
accounts payable / total assets
dummy; one if paid-in equity is less than total equity
EBITDA / total liabilities
log(age in years)
log(total assets)
public taxes payable / total assets
total equity / total assets
working capital / total revenues

Figure IA.1: AUC obtained when using the different input variable sets



The AUC values are the mean of AUC across the eight evaluations for the years 2010-2020. For FSR, BSR, GEN, LAS, FIL, and RAN, we let the predetermined number of input variables in the sets vary between 1 and 15. Moreover, for RAN, we report the mean value and the standard deviation range (mean $\pm$ one standard deviation). Top and bottom panels show in-sample AUC and out-of-sample AUC, respectively. Left and right panels display results when using LR and DNN, respectively, as estimation technique.

Figure IA.2: The percentage marginal increase in AP



The percentage marginal increase in in-sample (left panel) and out-of-sample (right panel) AP when expanding the size of the input variable set selected by the LASSO method. The AP values represent the mean of AP across the 11 evaluations for the years 2010-2020.

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